A POINT OF A SITE AS A MORPHISM OF SITES FROM THE SITE OF SETS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The site associated to a powerset. Same chapter, Remark remark-pt-topos. Supplies the site whose topos is equivalent to the topos of a point and the description of the equivalence.
- The definition of a point of a site and of a morphism of sites. Same chapter, Definition definition-point and Definition definition-morphism-sites. Supply the conditions verified for the functor.
- The pushforward along a point. Same chapter, Equation equation-skyscraper. Identifies the pushforward of a set with the maps out of the values of the functor.

Source reference. Original remark 00XD: sources/stacks-sites/pinned-original-source.tex, source lines 8344–8344. Navigation pointer invoked at proof line 8344; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00YA: sources/stacks-sites/pinned-original-source.tex, source lines 8358–8358. Navigation pointer invoked at proof line 8358; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original definition 00Y5: sources/stacks-sites/pinned-original-source.tex, source lines 8278–8278. Navigation pointer invoked at proof line 8278; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 00X1: sources/stacks-sites/pinned-original-source.tex, source lines 8371–8371. Navigation pointer invoked at proof line 8371; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 04EI: sources/stacks-sites/pinned-original-source.tex, source lines 8377–8377. Navigation pointer invoked at proof line 8377; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let p be a point of $\mathcal{C}$ given by $u : \mathcal{C} \to Sets$. Let S_0 be an infinite set such that $u(U) \subset S_0$ for all $U \in \mathrm{Ob}(\mathcal{C})$. Let $\mathcal{S}$ be the site constructed out of the powerset $S = \mathcal{P}(S_0)$ in Remark 00XD. Then*

1. *there is an equivalence $i : Sh(pt) \to Sh(\mathcal{S})$,*
2. *the functor $u : \mathcal{C} \to \mathcal{S}$ induces a morphism of sites $f : \mathcal{S} \to \mathcal{C}$, and*
3. *the composition*
$$Sh(pt) \to Sh(\mathcal{S}) \to Sh(\mathcal{C})$$
is the morphism of topoi (p_*, p^{-1}) of Lemma 00YA.

Proof. Part (1) we saw in Remark 00XD. Moreover, recall that the equivalence associates to the set E the sheaf i_*E on $\mathcal{S}$ defined by the rule $V \mapsto \mathrm{Mor}_{Sets}(V, E)$. Part (2) is clear from the definition of a point of $\mathcal{C}$ (Definition 00Y5) and the definition of a morphism of sites (Definition 00X1). Finally, consider f_*i_*E. By construction we have
$$f_*i_*E(U) = i_*E(u(U)) = \mathrm{Mor}_{Sets}(u(U), E)$$
which is equal to $p_*E(U)$, see Equation (04EI). This proves (3). □